MIDTERM TASK 3 – Normalization using Power Query

This task focuses on organizing and transforming data into a more structured and efficient format using Power Query. Students will apply normalization techniques by separating repeating data into related tables, removing redundancy, and ensuring the dataset follows proper data organization principles. The objective is to create a clean, well-structured dataset that improves data consistency and makes analysis easier.

Data Cleaning Start

trans_id	Date	client_id	Contact	department_id	state_id	payment_id	Revenue	Profit	Profit Margin
t-101	5/30/2023	AMAZON.COM, INC.	Bill Smith	Cloud Tech	Texas	Transfer	$ 4,500.00	$ 598.00	13.3%
t-102	5/30/2023	TESLA, INC.	Ken Singh	Strategy	New York	PayPal	$ 3,800.00	$ 1,045.00	27.5%
t-103	5/30/2023	NETFLIX, INC.	Harley Fritz	Strategy	New York	Others	$ 3,712.50	$ 1,009.00	27.2%
t-104	5/30/2023	THE PROCTER & GAMBLE COMPANY	Nyla Novak	Operations	Florida	Others	$ 4,994.22	$ 779.00	15.6%
t-105	5/30/2023	THE GOLDMAN SACHS GROUP, INC.	David Rasmussen	Operations	Florida	Check	$ 5,000.00	$ 684.00	13.7%
t-106	5/31/2023	JPMORGAN CHASE & CO.	Ivan Hiney	Cloud Tech	Texas	Transfer	$ 6,100.00	$ 544.00	8.9%
t-107	5/31/2023	MORGAN STANLEY	Jonha Ma	Cloud Tech	Texas	Transfer	$ 4,625.00	$ 670.00	14.5%
t-108	5/31/2023	CITIGROUP INC.	Jordan Boone	Cloud Tech	Texas	Transfer	$ 3,800.00	$ 2,045.00	53.8%
t-109	5/31/2023	BANK OF AMERICA CORPORATION	Kylee Townsend	Cloud Tech	Texas	Card	$ 3,600.00	$ 1,564.00	43.4%
t-110	5/31/2023	WALMART INC.	Nora Rollins	Cloud Tech	Texas	Check	$ 5,100.00	$ 1,220.00	23.9%
t-111	5/31/2023	TARGET CORPORATION	Brendan Wallace	Cloud Tech	Texas	Check	$ 4,750.00	$ 1,435.00	30.2%
t-112	5/31/2023	COSTCO WHOLESALE CORPORATION	Conor Wise	Operations	Florida	Transfer	$ 6,000.00	$ 998.00	16.6%
t-113	5/31/2023	MCDONALD'S CORPORATION	Steven Michael	Big Data	California	Check	$ 4,500.00	$ 780.00	17.3%
t-114	6/1/2023	EXXON MOBIL CORPORATION	Lucia Mckay	Big Data	California	Card	$ 4,994.22	$ 1,044.00	20.9%
t-115	6/1/2023	VERIZON COMMUNICATIONS INC.	Jose Roach	Big Data	California	Transfer	$ 3,712.50	$ 1,222.00	32.9%
t-116	6/1/2023	THE HOME DEPOT, INC.	Franklin Wrigt	Big Data	California	Transfer	$ 4,950.00	$ 1,065.00	21.5%
t-117	6/1/2023	CISCO SYSTEMS, INC.	Alia Thornton	Operations	Florida	Transfer	$ 4,750.00	$ 810.00	17.1%
t-118	6/1/2023	CHEVRON CORPORATION	Denzel Flores	Operations	Florida	Transfer	$ 7,320.00	$ 933.00	12.7%
t-119	6/1/2023	AT&T INC.	Bruno Cordova	Big Data	California	Transfer	$ 5,087.50	$ 655.00	12.9%
t-120	6/1/2023	INTEL CORPORATION	Jaylynn Napp	Big Data	California	Transfer	$ 4,500.00	$ 722.00	16.0%
t-121	6/1/2023	GENERAL MOTORS COMPANY	Bruce Rich	Big Data	California	Card	$ 4,250.00	$ 901.00	21.2%
t-122	6/2/2023	MICROSOFT CORPORATION	Arturo Moore	Big Data	California	PayPal	$ 5,250.00	$ 1,349.00	25.7%
t-123	6/2/2023	COMCAST CORPORATION	Bryce Carpenter	Strategy	New York	PayPal	$ 6,500.00	$ 1,288.00	19.8%
t-124	6/2/2023	DELL TECHNOLOGIES INC.	Jaidyn Andersen	Strategy	New York	PayPal	$ 7,500.00	$ 1,664.00	22.2%
t-125	6/2/2023	JOHNSON & JOHNSON	Mark Walm	Strategy	New York	Transfer	$ 5,500.00	$ 1,320.00	24.0%
t-126	6/2/2023	FEDEX CORPORATION	Harry Lee	Strategy	New York	Transfer	$ 4,625.00	$ 1,001.00	21.6%
t-127	6/2/2023	GENERAL ELECTRIC COMPANY	Josh Johnson	Strategy	New York	Transfer	$ 4,500.00	$ 960.00	21.3%
t-128	6/2/2023	LOCKHEED MARTIN CORPORATION	Mik Naam	Strategy	New York	Card	$ 5,400.00	$ 540.00	10.0%

Data Cleaning Start | VLOOKUP | REVENUE_FACT_COPY | REVENUE_FACT | Sheet2 | Sheet1 | CLIENT_DIM | STATE_DIM | PAYMENT_DIM | DEPARTMENT_DIM

vlookup

trans_id	Date	client_id	client_id	client	Contact	department_id	department_id	department	state_id	state	payment_id	payment	Revenue	Profit	Profit Margin
t-101	5/30/2023	c-101	c-101	AMAZON.COM, INC.	Bill Smith	d-102	d-102	Cloud Tech	s-104	Texas	p-105	Transfer	$ 4,500.00	$ 598.00	13.3%
t-102	5/30/2023	c-123	c-123	TESLA, INC.	Ken Singh	d-104	d-104	Strategy	s-103	New York	p-104	PayPal	$ 3,800.00	$ 1,045.00	27.5%
t-103	5/30/2023	c-121	c-121	NETFLIX, INC.	Harley Fritz	d-104	d-104	Strategy	s-103	New York	p-103	Others	$ 3,712.50	$ 1,009.00	27.2%
t-104	5/30/2023	c-126	c-126	THE PROCTER & GAMBLE COMPANY	Nyla Novak	d-103	d-103	Operations	s-102	Florida	p-103	Others	$ 4,994.22	$ 779.00	15.6%
t-105	5/30/2023	c-124	c-124	THE GOLDMAN SACHS GROUP, INC.	David Rasmussen	d-103	d-103	Operations	s-102	Florida	p-102	Check	$ 5,000.00	$ 684.00	13.7%
t-106	5/31/2023	c-116	c-116	JPMORGAN CHASE & CO.	Ivan Hiney	d-102	d-102	Cloud Tech	s-104	Texas	p-105	Transfer	$ 6,100.00	$ 544.00	8.9%
t-107	5/31/2023	c-120	c-120	MORGAN STANLEY	Jonha Ma	d-102	d-102	Cloud Tech	s-104	Texas	p-105	Transfer	$ 4,625.00	$ 670.00	14.5%
t-108	5/31/2023	c-106	c-106	CITIGROUP INC.	Jordan Boone	d-102	d-102	Cloud Tech	s-104	Texas	p-105	Transfer	$ 3,800.00	$ 2,045.00	53.8%
t-109	5/31/2023	c-103	c-103	BANK OF AMERICA CORPORATION	Kylee Townsend	d-102	d-102	Cloud Tech	s-104	Texas	p-101	Card	$ 3,600.00	$ 1,564.00	43.4%
t-110	5/31/2023	c-128	c-128	WALMART INC.	Nora Rollins	d-102	d-102	Cloud Tech	s-104	Texas	p-102	Check	$ 5,100.00	$ 1,220.00	23.9%
t-111	5/31/2023	c-122	c-122	TARGET CORPORATION	Brendan Wallace	d-102	d-102	Cloud Tech	s-104	Texas	p-102	Check	$ 4,750.00	$ 1,435.00	30.2%
t-112	5/31/2023	c-108	c-108	COSTCO WHOLESALE CORPORATION	Conor Wise	d-103	d-103	Operations	s-102	Florida	p-105	Transfer	$ 6,000.00	$ 998.00	16.6%
t-113	5/31/2023	c-118	c-118	MCDONALD'S CORPORATION	Steven Michael	d-101	d-101	Big Data	s-101	California	p-102	Check	$ 4,500.00	$ 780.00	17.3%
t-114	6/1/2023	c-110	c-110	EXXON MOBIL CORPORATION	Lucia Mckay	d-101	d-101	Big Data	s-101	California	p-101	Card	$ 4,994.22	$ 1,044.00	20.9%
t-115	6/1/2023	c-127	c-127	VERIZON COMMUNICATIONS INC.	Jose Roach	d-101	d-101	Big Data	s-101	California	p-105	Transfer	$ 3,712.50	$ 1,222.00	32.9%
t-116	6/1/2023	c-125	c-125	THE HOME DEPOT, INC.	Franklin Wrigt	d-101	d-101	Big Data	s-101	California	p-105	Transfer	$ 4,950.00	$ 1,065.00	21.5%
t-117	6/1/2023	c-105	c-105	CISCO SYSTEMS, INC.	Alia Thornton	d-103	d-103	Operations	s-102	Florida	p-105	Transfer	$ 4,750.00	$ 810.00	17.1%
t-118	6/1/2023	c-104	c-104	CHEVRON CORPORATION	Denzel Flores	d-103	d-103	Operations	s-102	Florida	p-105	Transfer	$ 7,320.00	$ 933.00	12.7%
t-119	6/1/2023	c-102	c-102	AT&T INC.	Bruno Cordova	d-101	d-101	Big Data	s-101	California	p-105	Transfer	$ 5,087.50	$ 655.00	12.9%
t-120	6/1/2023	c-114	c-114	INTEL CORPORATION	Jaylynn Napp	d-101	d-101	Big Data	s-101	California	p-105	Transfer	$ 4,500.00	$ 722.00	16.0%
t-121	6/1/2023	c-113	c-113	GENERAL MOTORS COMPANY	Bruce Rich	d-101	d-101	Big Data	s-101	California	p-101	Card	$ 4,250.00	$ 901.00	21.2%
t-122	6/2/2023	c-119	c-119	MICROSOFT CORPORATION	Arturo Moore	d-101	d-101	Big Data	s-101	California	p-104	PayPal	$ 5,250.00	$ 1,349.00	25.7%
t-123	6/2/2023	c-107	c-107	COMCAST CORPORATION	Bryce Carpenter	d-104	d-104	Strategy	s-103	New York	p-104	PayPal	$ 6,500.00	$ 1,288.00	19.8%
t-124	6/2/2023	c-109	c-109	DELL TECHNOLOGIES INC.	Jaidyn Andersen	d-104	d-104	Strategy	s-103	New York	p-104	PayPal	$ 7,500.00	$ 1,664.00	22.2%
t-125	6/2/2023	c-115	c-115	JOHNSON & JOHNSON	Mark Walm	d-104	d-104	Strategy	s-103	New York	p-105	Transfer	$ 5,500.00	$ 1,320.00	24.0%
t-126	6/2/2023	c-111	c-111	FEDEX CORPORATION	Harry Lee	d-104	d-104	Strategy	s-103	New York	p-105	Transfer	$ 4,625.00	$ 1,001.00	21.6%
t-127	6/2/2023	c-112	c-112	GENERAL ELECTRIC COMPANY	Josh Johnson	d-104	d-104	Strategy	s-103	New York	p-105	Transfer	$ 4,500.00	$ 960.00	21.3%
t-128	6/2/2023	c-117	c-117	LOCKHEED MARTIN CORPORATION	Mik Naam	d-104	d-104	Strategy	s-103	New York	p-101	Card	$ 5,400.00	$ 540.00	10.0%

Data Cleaning Start | VLOOKUP | REVENUE_FACT_COPY | REVENUE_FACT | Sheet2 | Sheet1 | CLIENT_DIM | STATE_DIM | PAYMENT_DIM | DEPARTMENT_DIM

Revenue fact copy

trans_id	Date	client_id	department_id	state_id	payment_id	Revenue	Profit	Profit Margin
t-101	5/30/2023	c-101	d-102	s-104	p-105	$ 4,500.00	$ 598.00	13.3%
t-102	5/30/2023	c-123	d-104	s-103	p-104	$ 3,800.00	$ 1,045.00	27.5%
t-103	5/30/2023	c-121	d-104	s-103	p-103	$ 3,712.50	$ 1,009.00	27.2%
t-104	5/30/2023	c-126	d-103	s-102	p-103	$ 4,994.22	$ 779.00	15.6%
t-105	5/30/2023	c-124	d-103	s-102	p-102	$ 5,000.00	$ 684.00	13.7%
t-106	5/31/2023	c-116	d-102	s-104	p-105	$ 6,100.00	$ 544.00	8.9%
t-107	5/31/2023	c-120	d-102	s-104	p-105	$ 4,625.00	$ 670.00	14.5%
t-108	5/31/2023	c-106	d-102	s-104	p-105	$ 3,800.00	$ 2,045.00	53.8%
t-109	5/31/2023	c-103	d-102	s-104	p-101	$ 3,600.00	$ 1,564.00	43.4%
t-110	5/31/2023	c-128	d-102	s-104	p-102	$ 5,100.00	$ 1,220.00	23.9%
t-111	5/31/2023	c-122	d-102	s-104	p-102	$ 4,750.00	$ 1,435.00	30.2%
t-112	5/31/2023	c-108	d-103	s-102	p-105	$ 6,000.00	$ 998.00	16.6%
t-113	5/31/2023	c-118	d-101	s-101	p-102	$ 4,500.00	$ 780.00	17.3%
t-114	6/1/2023	c-110	d-101	s-101	p-101	$ 4,994.22	$ 1,044.00	20.9%
t-115	6/1/2023	c-127	d-101	s-101	p-105	$ 3,712.50	$ 1,222.00	32.9%
t-116	6/1/2023	c-125	d-101	s-101	p-105	$ 4,950.00	$ 1,065.00	21.5%
t-117	6/1/2023	c-105	d-103	s-102	p-105	$ 4,750.00	$ 810.00	17.1%
t-118	6/1/2023	c-104	d-103	s-102	p-105	$ 7,320.00	$ 933.00	12.7%
t-119	6/1/2023	c-102	d-101	s-101	p-105	$ 5,087.50	$ 655.00	12.9%
t-120	6/1/2023	c-114	d-101	s-101	p-105	$ 4,500.00	$ 722.00	16.0%
t-121	6/1/2023	c-113	d-101	s-101	p-101	$ 4,250.00	$ 901.00	21.2%
t-122	6/2/2023	c-119	d-101	s-101	p-104	$ 5,250.00	$ 1,349.00	25.7%
t-123	6/2/2023	c-107	d-104	s-103	p-104	$ 6,500.00	$ 1,288.00	19.8%
t-124	6/2/2023	c-109	d-104	s-103	p-104	$ 7,500.00	$ 1,664.00	22.2%
t-125	6/2/2023	c-115	d-104	s-103	p-105	$ 5,500.00	$ 1,320.00	24.0%
t-126	6/2/2023	c-111	d-104	s-103	p-105	$ 4,625.00	$ 1,001.00	21.6%
t-127	6/2/2023	c-112	d-104	s-103	p-105	$ 4,500.00	$ 960.00	21.3%
t-128	6/2/2023	c-117	d-104	s-103	p-101	$ 5,400.00	$ 540.00	10.0%

Data Cleaning Start | VLOOKUP | REVENUE_FACT_COPY | REVENUE_FACT | Sheet2 | Sheet1 | CLIENT_DIM | STATE_DIM | PAYMENT_DIM | DEPARTMENT_DIM

Client dim table

client	client_id	Contact
AMAZON.COM, INC.	c-101	Bill Smith
AT&T INC.	c-102	Bruno Cordova
BANK OF AMERICA CORPORATION	c-103	Kylee Townsend
CHEVRON CORPORATION	c-104	Denzel Flores
CISCO SYSTEMS, INC.	c-105	Alia Thornton
CITIGROUP INC.	c-106	Jordan Boone
COMCAST CORPORATION	c-107	Bryce Carpenter
COSTCO WHOLESALE CORPORATION	c-108	Conor Wise
DELL TECHNOLOGIES INC.	c-109	Jaidyn Andersen
EXXON MOBIL CORPORATION	c-110	Lucia Mckay
FEDEX CORPORATION	c-111	Harry Lee
GENERAL ELECTRIC COMPANY	c-112	Josh Johnson
GENERAL MOTORS COMPANY	c-113	Bruce Rich
INTEL CORPORATION	c-114	Jaylynn Napp
JOHNSON & JOHNSON	c-115	Mark Walm
JPMORGAN CHASE & CO.	c-116	Ivan Hiney
LOCKHEED MARTIN CORPORATION	c-117	Mik Naam
MCDONALD'S CORPORATION	c-118	Steven Michael
MICROSOFT CORPORATION	c-119	Arturo Moore
MORGAN STANLEY	c-120	Jonha Ma
NETFLIX, INC.	c-121	Harley Fritz
TARGET CORPORATION	c-122	Brendan Wallace
TESLA, INC.	c-123	Ken Singh
THE GOLDMAN SACHS GROUP, INC.	c-124	David Rasmussen
THE HOME DEPOT, INC.	c-125	Franklin Wrigt
THE PROCTER & GAMBLE COMPANY	c-126	Nyla Novak
VERIZON COMMUNICATIONS INC.	c-127	Jose Roach
WALMART INC.	c-128	Nora Rollins

ning Start | VLOOKUP | REVENUE_FACT_COPY | REVENUE_FACT | Sheet2 | Sheet1 | CLIENT_DIM | STATE_DIM | PAYM

State dim table

State	State_id
California	s-101
Florida	s-102
New York	s-103
Texas	s-104

Payment dim table

payment	payment_id
Card	p-101
Check	p-102
Others	p-103
PayPal	p-104
Transfer	p-105

Department dim table

department	department_id
Big Data	d-101
Cloud Tech	d-102
Operations	d-103
Strategy	d-104
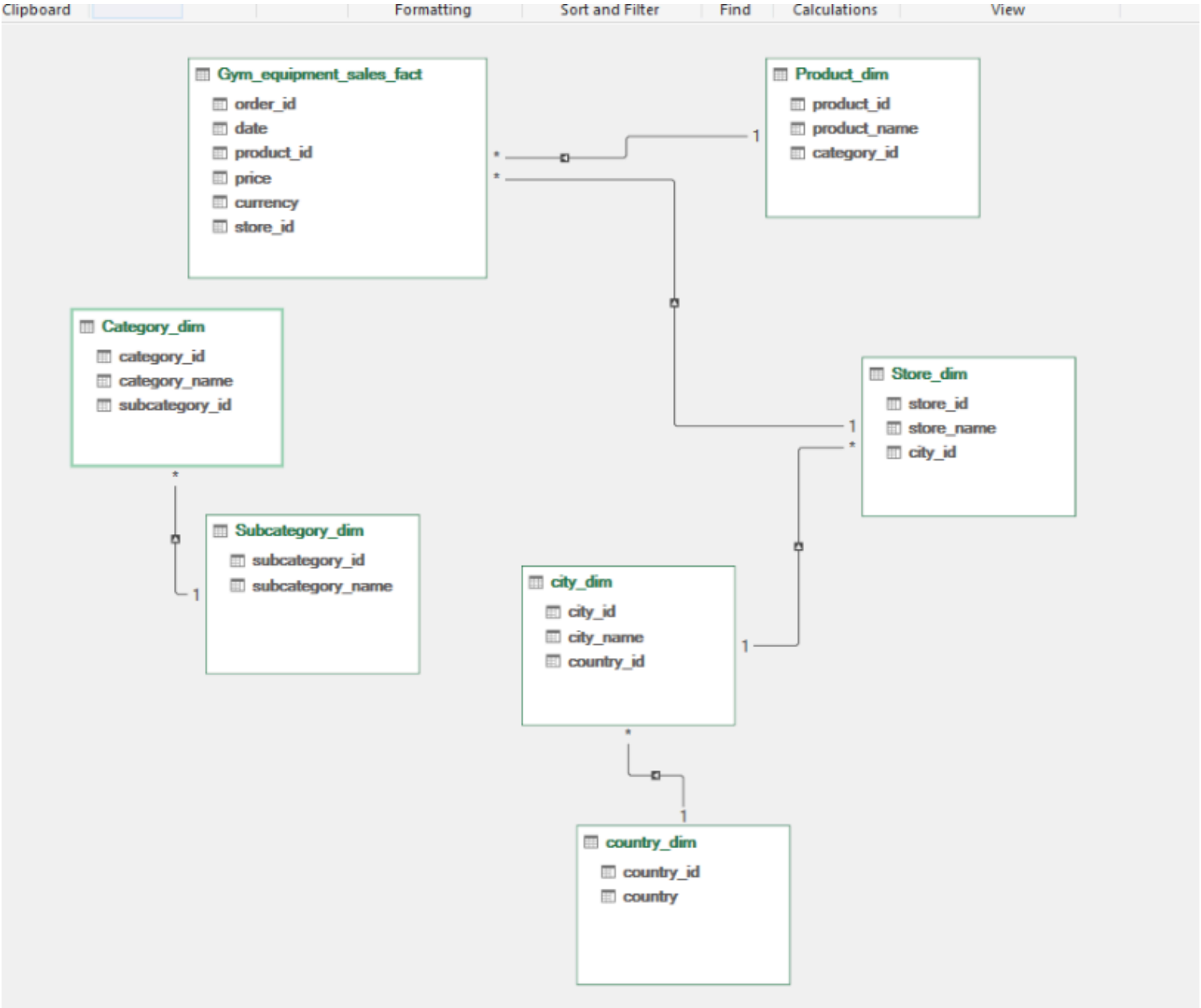
Table relationships diagram